

FIITJEE –Phase Test (JEE-Advanced)**PHYSICS, CHEMISTRY & MATHEMATICS**

Pattern - CPT-2

QP CODE:**PAPER - 1**

Time Allotted: 3 Hours

Maximum Marks: 186

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A & B** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-04)** – Contains Six (04) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **–1 marks** for wrong answer.
- (ii) **PART–A (05–12)** contains (8) Multiple Choice Questions which have **One or More Than One Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –1 In all other cases.
 For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–1 marks**, as a wrong option is also darkened.
- (iii) **Part-B (01-06)** contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

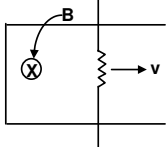
Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCH – ABC LOT NAO PAPER-2123 (PHASE–5)

SECTION – I : PHYSICS**PART – A: (Single Correct Answer Type)**

This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

- In an oscillating LC circuit the maximum charge on the capacitor is Q . The charge on the capacitor when the energy is stored equally between the electric and magnetic field is
 (A) $Q/2$ (B) $Q/\sqrt{3}$
 (C) $Q/\sqrt{2}$ (D) Q
- A conducting bar is pulled with a constant speed v on a smooth conducting rail. The region has a steady magnetic field of induction B as shown in the figure. If the speed of the bar is doubled then the rate of heat dissipation will
 (A) remain constant (B) become quarter of the initial value
 (C) become four times (D) get doubled
 

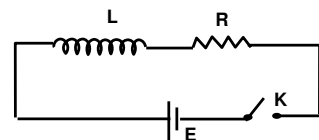
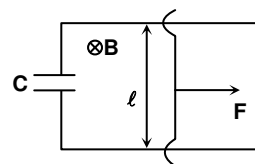
The diagram shows a rectangular circuit. A vertical rod on the right is being pulled to the right with velocity v . It is connected to a horizontal rail on the left and another horizontal rail on the right. A magnetic field B is indicated by a circle with a cross, pointing into the page.
- A prism of refractive index $\sqrt{2}$ refracting angle A produces minimum deviation D_m of a ray on one face at an angle of incidence 45° . The values of A and D_m are respectively.
 (A) $45^\circ, 45^\circ$ (B) $45^\circ, 60^\circ$
 (C) $60^\circ, 30^\circ$ (D) $60^\circ, 45^\circ$
- The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is:
 (A) 6% (B) 2.5%
 (C) 3.5% (D) 4.5%

Space for rough work

PART – A: (Multi Correct Answer Type)

This section contains **08 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

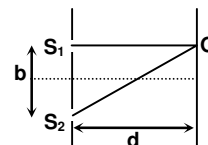
5. A circular conducting loop is placed in a uniform magnetic field with its plane perpendicular to the field. An emf is induced in the loop if
 (A) it is translated (B) it is rotated about its axis
 (C) it is rotated about a diameter (D) it is deformed
6. An inductor of inductance 2.0 mH is connected across a charged capacitor of capacitance $5.0 \mu\text{F}$, and the resulting LC circuit is set oscillating at its natural frequency. Let Q denote the instantaneous charge on the capacitor, and I the current in the circuit. It is found that the maximum value of Q is $200 \mu\text{C}$.
 (A) When $Q = 200 \mu\text{C}$, the value of I is zero
 (B) The maximum value of I is 2 Ampere .
 (C) When $Q = 200 \mu\text{C}$, the value of I is 2 Ampere
 (D) The maximum value of I is 1 Ampere
7. A conducting wire of length ℓ and mass m can slide without friction on two parallel rails and is connected to capacitance C . Whole system lies in a magnetic field B and a constant force F is applied to the rod. Then
 (A) the rod moves with constant velocity
 (B) the rod moves with an acceleration of $\frac{F}{m + B^2 \ell^2 c}$
 (C) there is constant charge on the capacitor.
 (D) charge on the capacitor increases with time
8. Shown in the figure is a R-L circuit. Just after the Key (K) is closed
 (A) the current in the circuit is zero
 (B) potential drop across the resistor is zero.
 (C) emf developed across the inductor equals the emf of the battery
 (D) no heat is dissipated in the circuit.



Space for rough work

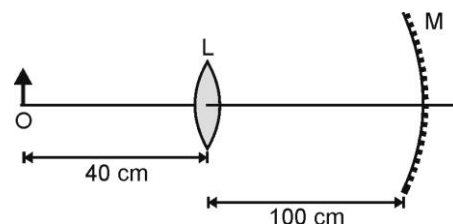
9. The mass number of a nucleus is
 (A) May be equal to its atomic number.
 (B) Always less than its atomic number.
 (C) May be more than its atomic number
 (D) May be more than or may be equal to its atomic number.
10. The potential difference applied to an X-ray tube is increased. As a result in the emitted radiation.
 (A) The intensity increases. (B) The minimum wavelength increases.
 (C) The intensity remains unchanged. (D) The minimum wavelength decreases.

11. White light is used to illuminate the two slits in a Young's double slit experiment. The separation between the slits is b and the screen is at a distance $D (>> b)$ from the slits. At a point on the screen directly in front of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are



- (A) $\lambda = \frac{b^2}{D}$ (B) $\lambda = \frac{2b^2}{D}$
 (C) $\lambda = \frac{b^2}{3D}$ (D) $\lambda = \frac{b^2}{5D}$

12. An object is placed at a distance 40 cm from a convex lens of focal length = 20 cm. A concave mirror of focal length 30 cm is also placed on other side of lens at a distance 100 cm. Final image of object is formed through lens after reflection from mirror ; then
 (A) The final image is real, erect and of same size of object
 (B) The final image is real, inverted and of same size of object
 (C) Distance between final image and mirror is 60 cm
 (D) Distance between final image and lens is 40 cm



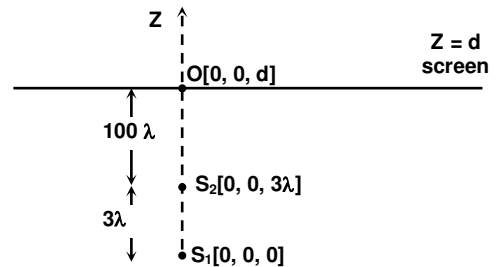
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PART – B: (Integer Answer Type)

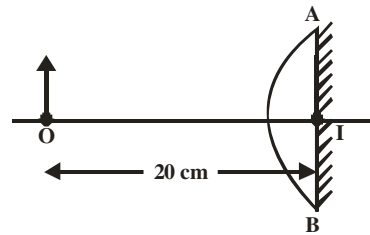
This section contains **06 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer and **there will be no negative marking**.

1. The wavelength of the first line of Balmer series of hydrogen atom is λ . Then the wavelength of the corresponding line (for the same transition) of a double ionised lithium atom ($z = 3$) is λ/k . (Where k is in integer). Find k .

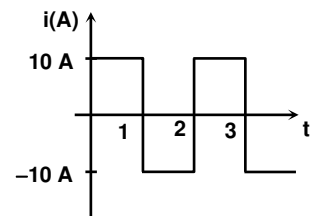
2. Two coherent point sources S_1 and S_2 of monochromatic light of wavelength $\lambda = 4500 \text{ \AA}$ are located as shown in the figure [$S_1S_2 = 3\lambda$]. A screen which can be represented by plane $z = d$. Then, find the number of dark circular fringes formed on the screen.



3. A point object is placed at a distance of 20 cm from a thin plano-convex lens of focal length 15 cm. If the plane surface is silvered, the image will form at $3N$ cm left of AB. Find the value of N .



4. The figure shows an alternating current. The magnitude of direct current which would produce the same heat in an identical resistance in the same time as produced by the given A-C is $2K$ ampere. Find the value of K .



5. A wire of length $\ell = 6 \pm 0.06$ cm and radius $r = 0.5 \pm 0.005$ cm and mass $m = 0.3 \pm 0.003$ gm. Then find the maximum percentage error in density.
6. The coefficient of mutual inductance of the two coils is 0.5 H. If the current is increased from 2 to 3 A in 0.01 sec. in one of them, then the induced e.m.f. in the second coil is $25K$ Volt. Find the value of K .

Space for rough work

SECTION – II : CHEMISTRY**PART – A: (Single Correct Answer Type)**

This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

- Electrolysis of dil. H_2SO_4 generates $\text{H}_2\text{S}_2\text{O}_8$ (per disulphuric acid). What current is needed to produce $\text{H}_2\text{S}_2\text{O}_8$ at the rate of 1 mole per hour assuming 75% charge efficiency.
(A) 71.48 amp. (B) 17.48 amp.
(C) 35.74 amp. (D) 53.74 amp.
- 1.3245 g of a monobasic acid when dissolved in 100 g of water lowers the freezing point by 0.2046°C . 0.2 g of the same acid when dissolved and titrated required 15.1 ml of $\frac{N}{10}$ alkali. (Assuming molarity = molality), the pH of acid solution is: (K_f of water $1.86 \text{ K kg mol}^{-1}$)
(A) 1 (B) 2
(C) 3 (D) 4
- The distance between tetrahedral and octahedral void in the fcc lattice will be (given edge length of unit cell = a)
(A) $\frac{\sqrt{3}a}{4}$ (B) $\sqrt{3}a$
(C) $\sqrt{3}\frac{a}{2}$ (D) $\frac{a}{\sqrt{3}}$
- What is obtained when mercuric chloride is heated with SO_2 ?
(A) Hg (B) Hg_2Cl_2
(C) HgSO_4 (D) Hg_2O

PART – A: (Multi Correct Answer Type)

This section contains **08 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- If electrolysis of 1 M NiSO_4 is carried with inert electrodes at pH = 7 at 25°C (the pressure of gases are taken 1 bar).
($E^\circ_{\text{Ni}^{2+}/\text{Ni}} = -0.25 \text{ V}$ $E^\circ_{\text{H}_2\text{O}/\text{H}_2} = -0.83 \text{ V}$)
The product of electrolysis at cathode and at anode (initially) are:
(A) Ni at cathode (B) H_2 at cathode
(C) O_2 at anode (D) $\text{S}_2\text{O}_8^{2-}$ at anode
- Which of the following statements are correct:
(A) The flocculating value of PO_4^{3-} is less than that of SO_4^{2-} .
(B) Charcoal adsorbs gases with higher critical temperature more than gases with lower critical temperature.
(C) Chemisorption is multilayered.
(D) In the manufacture of NH_3 by Haber's process, molybdenum is used as promoter for the catalyst iron.

Space for rough work

- Yashpatil TG - @bohring_bot
7. The volume of which of the following unit cells will be independent of α, β, γ (interfacial angles)
- (A) Orthorhombic (B) Monoclinic
(C) Hexagonal (D) Rhombohedral
8. Among the following statements which is/are correct
- (A) Calamine and Siderite are carbonate ores.
(B) Argenite and Cuprites are oxide ores.
(C) Malachite and Azurites are ores of Cu.
(D) Carnallite and Sylvite are chloride ores.
9. Two liquids A and B form an ideal solution. The solution has a vapor pressure of 700 Torr at 80°C . It is distilled till $2/3^{\text{rd}}$ of the solution is collected as condensate. The composition of the condensate is $x'_A = 0.75$ and that of the residue is $x''_A = 0.30$. If the vapor pressure of the residue at 80°C is 600 Torr, which of the following is/are true?
- (A) The composition of the original liquid was $x_A = 0.6$.
(B) The composition of the original liquid was $x_A = 0.4$.
(C) $P_A^0 = 2500/3 \text{ Torr}$.
(D) $P_B^0 = 500 \text{ Torr}$.
10. Which of the following statement(s) is/are incorrect?
- (A) Manganese salts give a violet borax bead test in reducing flame.
(B) From mixed salt of AgCl and AgI, NH_4OH solution dissolves only AgCl.
(C) Fe^{3+} ions given a deep green precipitate, on adding $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution.
(D) On strong heating $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$, a polysulphide is resulted.
11. Which of the following statements is/are correct for the spinel structure of MgAl_2O_4 ?
- (A) 50% of octahedral void are occupied by ions
(B) Al^{3+} is equally distributed in tetrahedral and octahedral voids
(C) Oxide ions occupy ccp lattice
(D) 12.5% of tetrahedral voids are occupied by ions
12. Phenol is used as
- (A) an antiseptic (B) an insecticide
(C) a disinfectant (D) styptic
-

Space for rough work

PART – B: (Integer Answer Type)

*This section contains **06 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer and **there will be no negative marking**.*

1. A crystalline solid of a pure substance has a face-centred cubic structure with a cell edge of 400 pm. If the density of the substance in the crystal is 8 g cm^{-3} , then the number of atoms present in 256 g of the crystal is $N \times 10^{24}$. The value of N is.
 2. How many of following salt are first soluble in $\text{Na}_2\text{S}_2\text{O}_3$ solution, followed by a white ppt. formation and finally ppt. turns dark black on standing?
 AgNO_3 , $\text{Pb}(\text{CH}_3\text{COO})_2$, $\text{Hg}(\text{NO}_3)_2$, BiCl_3 , CuSO_4
 3. The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is.
 4. The depression in freezing point of 0.1 molal aq. NH_3 solution is 0.19 K_b value of NH_3 is 4.72×10^{-x} . Then x is... (Given K_f of H_2O is $1.86 \text{ kg/mol}^{-1}/\text{K}$)?
 5. 0.05 Faraday charge is passed in 100 ml, 0.1 M NaCl solution. Calculate the final pOH of the solution. Assume volume remains constant.
 6. How many gases/vapours are involved either as reactants or products in Serpeck's process used for concentration of bauxite?
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Space for rough work

SECTION – III : MATHEMATICS**PART – A: (Single Correct Answer Type)**

This section contains **04 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

- If p, q, r are in A.P., then the determinants $\begin{vmatrix} a^2 + 2^{n+1} + 2p & b^2 + 2^{n+2} + 3q & c^2 + p \\ 2^n + p & 2^{n+1} + q & 2q \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix}$ is equal to

(A) 1
(B) 0
(C) $a^2b^2c^2 - 2^n$
(D) $(a^2 + b^2 + c^2) - 2^nq$
- Solution of the differential equation $(x + 2y^3) \frac{dy}{dx} = y$ is

(A) $x = y^2(c + y^2)$
(B) $x = y(c - y^2)$
(C) $x = 2y(c - y^2)$
(D) $x = y(c + y^2)$
- Different words are formed using the letters of word FIITJEE, then the probability that the word contains all consonants together, is

(A) $\frac{1}{6}$
(B) $\frac{1}{7}$
(C) $\frac{1}{4}$
(D) none of these
- Three unit vectors a, b, c are such that $a + b = c$, then $|a - b + c|$ is equal to

(A) $\sqrt{3}$
(B) $\sqrt{2}$
(C) 2
(D) 1

Space for rough work

PART – A: (Multi Correct Answer Type)

This section contains **08 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

5. If P and Q are two 3×3 non-identity matrices with real entries such that $PP^T = I$, $QQ^T = I$ (P^T , Q^T are transpose of matrices P and Q respectively) and $\det(PQ) < 0$, then which of the following is/are always correct (where $\det$ stands for determinant of a matrix)
- (A) $\det(P + Q) < 0$
 (B) $\det(P + Q) = 0$
 (C) atleast one of the $\det(P - I)$ or $\det(Q - I)$ will be zero
 (D) $\det(P^{-1} + Q^{-1}) = \det(P + Q)$
6. If $\Delta(x) = \begin{vmatrix} x^2 + 4x - 3 & 2x + 4 & 13 \\ 2x^2 + 5x - 9 & 4x + 5 & 26 \\ 8x^2 - 6x + 1 & 16x - 6 & 104 \end{vmatrix} = ax^3 + bx^2 + cx + d$, then
- (A) $a = 3$
 (B) $b = 0$
 (C) $c = 0$
 (D) none of these
7. The area enclosed by the parabola $ay = 3(a^2 - x^2)$ and the x-axis is
- (A) $2 \int_0^a \frac{3}{a} (a^2 - x^2) dx$ sq. units
 (B) $2 \int_0^a \frac{3}{a} (x^2 - a^2) dx$ sq. units
 (C) $6a^2$ sq. units
 (D) $4a^2$ sq. units
8. Solution of differential equation $(e^x + 1) y dy = (y + 1) e^x dx$ is
- (A) $(e^x + 1)(y + 1) = ce^y$
 (B) $(e^x + 1)|(y + 1)| = ce^y$
 (C) $(e^x + 1)(y + 1) = \pm ce^y$
 (D) None of these

Space for rough work

9. If $a, b, c, x, y, z \in \mathbb{R}$ and $\vec{r}_1 = a\hat{i} + b\hat{j} + c\hat{k}$, $\vec{r}_2 = x\hat{i} + y\hat{j} + z\hat{k}$ are two vectors such that $|\vec{r}_1| = 5$, $|\vec{r}_2| = 6$ and $\vec{r}_1 \cdot \vec{r}_2 = 30$, then
- (A) $\frac{x}{a} = \frac{y}{b}$ (B) $\frac{x}{a} = \frac{z}{c}$
 (C) $x + y + z = a + b + c$ (D) $\frac{a+b+c}{x+y+z} = \frac{5}{6}$
10. Which of the following is/are correct?
- (A) The point of intersection of $\overline{AB}$ and $\overline{CD}$ where $A(4, 7, 8)$, $B(-1, -2, 1)$, $C(2, 3, 4)$ and $D(1, 2, 5)$ is $\left(\frac{3}{2}, \frac{5}{2}, \frac{9}{2}\right)$
 (B) If (α, β, γ) be a point on the plane $2x + 6y + 15z = 7$ then the least value of $\alpha^2 + \beta^2 + 25\gamma^2$ is 7
 (C) The circumcentre of the triangle formed by the points $(3, 2, -5)$, $(-3, 8, -5)$ and $(-3, 2, 1)$ is the point $(-1, 4, -3)$
 (D) The length of intercept on the line $\frac{x+1}{-1} = \frac{y-12}{5} = \frac{z-7}{2}$ by the surface $11x^2 - 5y^2 + z^2 = 0$ is $\sqrt{30}$ units
11. If A and B are two events such that $P(A) = \frac{3}{4}$ and $P(B) = \frac{4}{5}$, then
- (A) $P(A \cup B) \geq \frac{4}{5}$ (B) $P(A' \cap B) \geq \frac{1}{2}$
 (C) $\frac{11}{20} \leq P(A \cap B) \leq \frac{3}{4}$ (D) $P(A \cap B') \geq \frac{1}{5}$
12. A ship is fitted with three engines E_1 , E_2 and E_3 . The engines function independently of each other with respective probabilities $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{4}$. For the ship to be operational at least two of its engines must function. Let X denote the event that the ship is operational and let X_1 , X_2 and X_3 denote respectively the events that the engines E_1 , E_2 and E_3 are functioning. Which of the following is(are) true?
- (A) $P[X_1^c | X] = \frac{3}{16}$
 (B) $P[\text{Exactly two engines of the ship are functioning} | X] = \frac{7}{8}$
 (C) $P[X | X_2] = \frac{5}{16}$
 (D) $P[X | X_1] = \frac{7}{16}$

Space for rough work

PART – B: (Integer Answer Type)

This section contains **06 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer and **there will be no negative marking**.

1. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to _____
2. If the least area bounded by the curves $y = x^2 - 4$ and $y = \lambda x + 12$ is equal to $\frac{\alpha}{\beta}$, then $\left[\frac{\alpha}{20\beta} \right]$ is equal to _____ (where $[.]$ denotes the greatest integer function)
3. The slope of the tangent to a curve $y = f(x)$ at $\{x, f(x)\}$ is $(2x + 1)$. If the curve passes through the point $(1, 2)$, then the area bounded by the curve, the x-axis and line $x = 1$ is _____
4. The equation of a curve whose slope at any point is thrice its abscissa and which passes through $(-1, -3)$ is $2y = \lambda(x^2 - 3)$, then the value of λ is _____
5. Let $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = a\hat{i} + b\hat{j} + c\hat{k}$, where $a, b, c \in \{-3, -2, -1, 0, 1, 2\}$ if the number of possible vectors $\vec{q}$ such that $\vec{p} \cdot \vec{q} = 0$ is R, then R is _____
6. The system of equations $ax + y + z = a - 1$, $x + ay + z = a - 1$, $x + y + az = a - 1$ has no solution, then $\frac{5|a|^2}{4}$ is _____

Space for rough work

FIITJEE INTERNAL TEST

PHYSICS, CHEMISTRY & MATHEMATICS

CPT-2

QP CODE:

PAPER-1

ANSWERS

PHYSICS (SECTION -I)

PART - A

- | | | | |
|------------|----------|-------------|---------------|
| 1. C | 2. C | 3. C | 4. D |
| 5. C, D | 6. A, B | 7. B, D | 8. A, B, C, D |
| 9. A, C, D | 10. C, D | 11. A, C, D | 12. B, D |

PART - B

- | | | | |
|------|------|------|------|
| 1. 9 | 2. 3 | 3. 4 | 4. 5 |
| 5. 4 | 6. 2 | | |

CHEMISTRY (SECTION -II)

PART - A

- | | | | |
|------------|------------|-------------|------------|
| 1. A | 2. B | 3. A | 4. B |
| 5. A, C | 6. A, B, D | 7. A, C | 8. A, C, D |
| 9. A, C, D | 10. A, C | 11. A, C, D | 12. A, C |

PART - B

- | | | | |
|------|------|------|------|
| 1. 2 | 2. 5 | 3. 6 | 4. 5 |
| 5. 1 | 6. 5 | | |

MATHEMATICS (SECTION-III)

PART - A

- | | | | |
|------------|-------------|-------------|------------|
| 1. B | 2. D | 3. B | 4. C |
| 5. B, C, D | 6. B, C | 7. A, D | 8. A, B, C |
| 9. A, B, D | 10. A, C, D | 11. A, C, D | 12. B, D |

PART - B

- | | | | |
|----------|---------|---------|---------|
| 1. 11.00 | 2. 4.00 | 3. 0.83 | 4. 3.00 |
| 5. 25.00 | 6. 5.00 | | |

SOLUTION**PHYSICS****PART – A**

$$1. \quad E = \frac{1}{2} \frac{Q^2}{C} \quad \dots(i)$$

$$\frac{E}{2} = \frac{1}{2} \frac{Q'^2}{C} \quad \dots(ii)$$

From equation (i) and (ii)

$$\frac{1}{2} \frac{Q^2}{2C} = \frac{1}{2} \frac{Q'^2}{C}$$

$$Q' = \frac{Q}{\sqrt{2}}$$

$$2. \quad I = \frac{\varepsilon}{R} = \frac{B\ell V}{R}$$

$$P = I^2 R$$

$$P = \left(\frac{B\ell V}{R} \right)^2 R$$

$$P \propto V^2$$

$$V \rightarrow 2V$$

$$P \rightarrow 4P$$

$$3. \quad r = A/2, u = \sqrt{2}, i = 45^\circ, A = ? \text{ and } D_{\min} = ?$$

$$i \sin 45^\circ = \sqrt{2} \sin r$$

$$r = 30^\circ$$

$$A = 2r \Rightarrow A = 60^\circ$$

$$D_{\min} = 2i - A = 2 \times 45^\circ - 60^\circ = 30^\circ$$

$$4. \quad V = a^3$$

$$D = m/V = m/a^3$$

$$\frac{\Delta D}{D} \times 100 = \frac{\Delta m}{m} \times 100 + \frac{3\Delta a}{a} \times 100 = 1.5 + 3 \times 1$$

$$\frac{\Delta D}{D} \times 100 = 45\%$$

$$5. \quad Q = BA \cos \theta$$

Q changes $\rightarrow \phi$ changes

A changes $\rightarrow \phi$ changes

$$6. \quad L = 2\text{mH}, C = 0.5 \mu\text{F}, Q_{\min} = 200 \mu\text{C}$$

$$U = \frac{1}{2} LI^2, \text{ if } U = 0 \text{ then } I = 0$$

For maximum value of I

$$U = \frac{1}{2} LI_{\max}^2 = \frac{Q_{\max}^2}{2C}$$

$$I_{\max}^2 = \frac{Q_{\max}^2}{LC}$$

$$I_{\max} = \frac{Q_{\max}}{\sqrt{LC}} = \frac{200 \times 10^{-6}}{\sqrt{2 \times 10^{-3} \times 0.5 \times 10^{-6}}} = 2\text{A}$$

7.

$$a = \frac{dV}{dt}$$

$$\text{emf}(\varepsilon) = \frac{d\phi}{dt} = \frac{d}{dt}(\mathbf{B} \cdot \mathbf{A}) = B\ell \times \frac{dx}{dt}$$

$$\varepsilon = B\ell V$$

$$q = CV$$

$$q = CB\ell V$$

$$I = \frac{dq}{dt} = CB\ell \frac{dV}{dt}$$

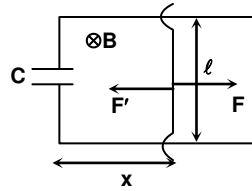
$$F' = IB\ell$$

$$\therefore F - F' = ma$$

$$F - IB\ell = ma$$

$$F - B^2 C \ell^2 a = ma$$

$$a = \frac{F}{M + B^2 \ell^2 C}$$



8.

$$i = i_0(1 - e^{-t/\tau}), \tau = L/R$$

$$(A) t = 0, i = 0$$

$$(B) i = 0, V = 0, V = IR \text{ and } E' = -E$$

$$(C) \text{ induced voltage of force applied emf}(f)$$

$$(D) \text{ Heat} = I^2 R t, I = 0, H = 0$$

9.

Basic concept

10.

No change in no. of electron if $V \uparrow$, $eV \uparrow$ and $\lambda \downarrow$

11.

$$\Delta x = \sqrt{b^2 + d^2} - d = \frac{n\lambda}{2} \quad (n = 1, 3, 5, \dots)$$

$$\left(\sqrt{b^2 + d^2}\right)^2 = \left(\frac{n\lambda}{2} + d\right)^2$$

$$b^2 + d^2 = \left(\frac{n\lambda}{2}\right)^2 + d^2 + n\lambda d$$

$$\lambda = \frac{b^2}{nd}$$

$$n = 1, \frac{b^2}{d}$$

$$n = 3, \frac{b^2}{3d}$$

$$n = 5, \frac{b^2}{5d}$$

12.

For lens,

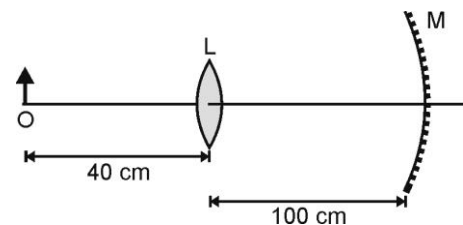
$$\frac{1}{v} + \frac{1}{40} = \frac{1}{20}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{40} = \frac{2-1}{40}$$

$$\Rightarrow v = 40 \text{ cm}$$

For mirror, ($u = -60$)

$$\frac{1}{v} + \frac{1}{-60} = \frac{1}{-30}$$



$$\frac{1}{v} = \frac{1}{-30} + \frac{1}{60}$$

$$\Rightarrow v = -60 \text{ cm}$$

Real, inverted and same size

PART – B

1. Blamer

$$n_2 = 3 \text{ to } \infty$$

$$n_1 = 2$$

$$\frac{1}{\lambda} = R(1)^2 \left(\frac{1}{22} - \frac{1}{32} \right) = R \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{5R}{36}$$

$$\text{Li}^{++}, z = 3, \quad \frac{1}{\lambda} = R(3)^2 \left[\frac{1}{22} - \frac{1}{32} \right] = 9 \times \frac{5R}{36}$$

$$\frac{1}{\lambda'} = 9 \frac{1}{\lambda}$$

$$\lambda' = \frac{\lambda}{9}$$

$$3. \quad P_{\text{total}} = P_\ell + P_m + P_\ell$$

$$= 2P_\ell + P_m$$

$$-\frac{1}{f_{\text{eq}}} = \frac{2}{f} + \left(\frac{1}{\infty} \right)$$

$$f_{\text{eq}} = -\frac{f}{2}$$

$$f_{\text{eq}} = -\frac{15}{2}$$

$$f = 15$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-20} = \frac{-2}{15}$$

$$\Rightarrow v = -12 = 3N$$

$$N = 4$$

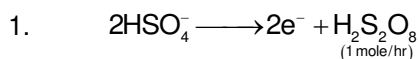
$$6. \quad \varepsilon_{\text{ind}} = \frac{M di}{dt} = \frac{M \Delta I}{\Delta t} = 0.5 \times \frac{(3-2)}{0.01}$$

$$\varepsilon_{\text{ind}} = 50 \text{ W} = 25 \text{ k}$$

$$\Rightarrow k = 2$$

CHEMISTRY

PART – A



2 mole/hr electrons need to be released = $2F$

$$= \frac{8}{3}F \text{ (assuming 75\% charge efficiency)}$$

$$= \frac{8}{3} \times 96500 \text{ C} = i \times 60 \times 60$$

$$i = 71.48 \text{ amp.}$$

2. Meq. of alkali = Meq. of acid

$$\Rightarrow 15.1 \times \frac{1}{10} = \frac{0.2}{M} \times 1000$$

On, solving $M = 132.45 \text{ g mol}^{-1}$

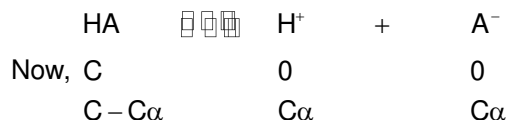
$$\text{Now, molality of acid soln (m)} = \frac{1.3245 \times 1000}{132.45 \times 100} = 0.1$$

$$\therefore \Delta T_f = i \times K_f \times m$$

$$0.2046 = i \times 1.86 \times 0.1$$

$$i = 1.1 = 1 + \alpha$$

$$\therefore \alpha = 0.1$$



$$[H^+] = C\alpha = 0.1 \times 0.1 = 10^{-2}$$

$$\therefore pH = 2$$

3. Octahedral void is at the centre of the cube and edge centre. Tetrahedral voids are present at $\frac{1}{4}$ th of the distance along each body diagonal.

4. When SO_2 is passed through $HgCl_2$ solution then due to lower reducing power of SO_2 , Hg^{2+} is reduced to Hg_2^{2+} .

Hence, (B) is correct while (A), (C) and (D) are wrong.

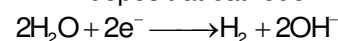
5. $H^+ + e^- \longrightarrow \frac{1}{2}H_2$ $E^\circ = 0.0 \text{ v}$

$$E = E_{H^+/H_2} = 0 - \frac{0.059}{1} \log \frac{1}{[H^+]}$$

$$= -0.059 \times 7 = -0.413$$

$$\text{So, } E_{H^+/H_2} < E_{Ni^{2+}/Ni}$$

Ni will deposit at cathode



$$E = -0.83 - \frac{0.059}{2} \log [OH^-]^2$$

$$= -0.83 + 0.059 \times 7$$

$$= -0.417$$

6. Chemisorption unimolecular layer.

8. Calamine – $ZnCO_3$, Siderite – $FeCO_3$, Argentite – Ag_2S , Cuprite – Cu_2O , Sylvite – KCl
Malachite – $CuCO_3 \cdot Cu(OH)_2$, Azurite – $Cu(OH)_2 \cdot 2CuCO_3$, Carnallite – $KCl \cdot MgCl_2 \cdot 6H_2O$

9. $x_A P_A^\circ + x_B P_B^\circ = 700$... (i)

$$x_A'' P_A^\circ + x_B'' P_B^\circ = 0.30 P_A^\circ + 0.7 P_B^\circ = 600 \quad \text{.. (ii)}$$

if moles of A & B initially are x & y then

$$x = 0.75 \times \frac{2}{3}(x+y) + 0.3 \times \frac{1}{3}(x+y)$$

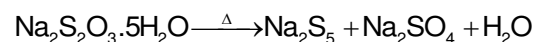
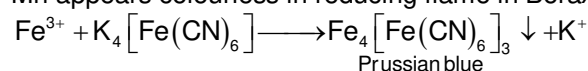
$$x_A = \frac{x}{x+y} \quad \text{or} \quad x_B = \frac{y}{x+y}$$

Solving given

$$x_A = 0.6 \quad x_B = 0.4$$

$$P_A^\circ = \frac{2500}{3} \text{ torr} \quad \& \quad P_A^\circ = 500 \text{ torr}$$

10. Mn appears colourless in reducing flame in Borax Bead Test.



11. For the spinel structure O^{2-} form fcc arrangement.

$$\text{No of } O^{2-} \text{ ions} = 4$$

$$\text{No of } Mg^{2+} \text{ ions} = \frac{1}{8} \times TV = \frac{1}{8} \times 8 = 1$$

$$\text{No of Al}^{3+} \text{ ions} = \frac{1}{2} \times \text{OV} = \frac{1}{2} \times 4 = 2$$

So 50% of octahedral voids are occupied and $\frac{1}{8}$ or 12.5% of tetrahedral voids are occupied.

PART – B

$$1. \quad d = \frac{Z \times M}{N_A \times a^3} \Rightarrow 8 = \frac{4 \times M}{6.022 \times 10^{23} \times (400 \times 10^{-10})^3}$$

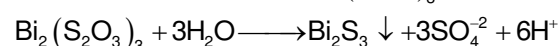
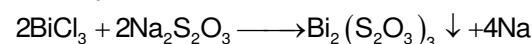
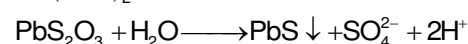
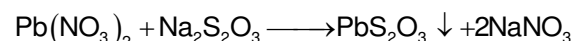
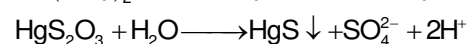
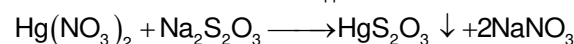
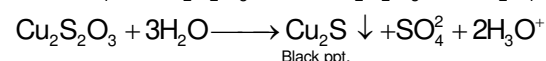
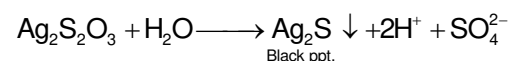
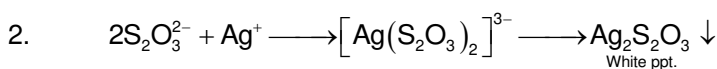
$$M = 76.8 \text{ g mol}^{-1}$$

$$76.8 \text{ g contain} = 6 \times 10^{23} \text{ atoms}$$

$$\therefore 256 \text{ g will contain} = 20 \times 10^{23} \text{ atoms}$$

$$= 2 \times 10^{24} \text{ atoms}$$

$$\therefore N = 2$$



$$4. \quad \Delta T_f = iK_f m$$

$$i = 1.0215$$

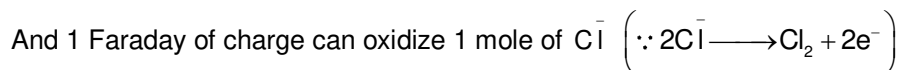


$$K_b = \frac{0.1 \times \alpha^2}{(1-\alpha)} = 4.72 \times 10^{-5}$$

$$\text{As } \alpha = (i-1)$$

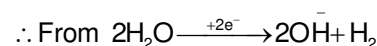
5. In given 100 ml 0.1 M NaCl solution,

$$\text{No. of moles of Cl}^- \text{ present} = \frac{100}{1000} \times 0.1 = 0.01$$



$\therefore$ Given 0.05 F is not utilized in electrolysis

Only 0.01 F will be utilized in electrolysis



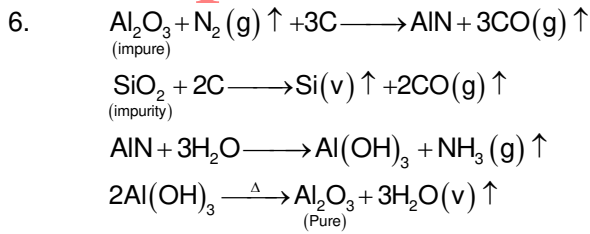
2 F of charge can produce 2 moles of OH^-

$\therefore$ 0.01 F will produce 0.01 mole of OH^-

$$\therefore \text{Molarity of } \text{OH}^- \text{ in solution} = \frac{0.01 \times 1000}{100} = 0.1 \text{ M}$$

$$\therefore \text{pOH} = -\log 10^{-1}$$

$$= 1$$



MATHEMATICS

Part – A

1. **B**

Applying $R_1 \rightarrow R_1 - R_3$ and $2q = p + r$

$$\begin{vmatrix} 2^{n+1} - 2^n + p & 2^{n+2} - 2^{n+1} + q & p + r \\ 2^n + p & 2^{n+1} + q & p + r \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix} = \begin{vmatrix} 2^n + p & 2^{n+1} + q & p + r \\ 2^n + p & 2^{n+1} + q & p + r \\ a^2 + 2^n + p & b^2 + 2^{n+1} + 2q & c^2 - r \end{vmatrix} = 0.$$

2. **D**

$$x dy + 2y^3 dy = y dx$$

$$\Rightarrow 2y dy = \frac{y dx - x dy}{y^2}$$

$$\Rightarrow x = y(c + y^2).$$

3. **B**

$$n(S) = \frac{7!}{(2!)^2}; n(A) = \frac{5! 3!}{(2!)^2}$$

$$\text{so probability } P(A) = \frac{5! 3!}{7!} = \frac{1}{7}.$$

4. **C**

Angle between a and b is $\frac{2\pi}{3}$ where $\vec{c}$ is along the angle bisector, $a - b$ is along another angle

bisector and hence perpendicular to c

$$|a - b| = \sqrt{3} \text{ hence } |a - b + c| = 2$$

5. **B, C, D**

$$P^T(P+Q)Q^T = (P+Q)^T \Rightarrow |P+Q| = 0, P^T(P-I) = -(P-I)^T$$

Either $|P| |P-I| = -|P-I|$ or $|Q| |Q-I| = -|Q-I|$, because $|P| |Q| < 0$

$$P(P^{-1} + Q^{-1})Q = P+Q \Rightarrow |P^{-1} + Q^{-1}| = -|P+Q| = 0$$

6. **B, C**

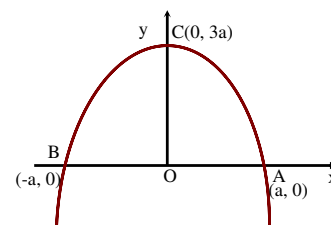
$$\Delta'(x) = \begin{vmatrix} 2x+4 & 2x+4 & 13 \\ 4x+5 & 4x+5 & 26 \\ 16x-6 & 16x-6 & 104 \end{vmatrix} + \begin{vmatrix} x^2+4x-3 & 2 & 13 \\ 2x^2+5x-9 & 4 & 26 \\ 8x^2-6x+1 & 16 & 104 \end{vmatrix} = 0 + 2 \times 13 \times (0) = 0.$$

$$\Rightarrow \Delta(x) = \text{constant} \Rightarrow a = 0, b = 0, c = 0.$$

7. **A, D**

The parabola cuts the x -axis at points $(-a, 0)$ and $(a, 0)$.

$$A = \int_{-a}^a y dx = 2 \frac{3}{a} \int_0^a (a^2 - x^2) dx$$



8. **A, B, C**

$$(e^x + 1) y dy = (y + 1) e^x dx \Rightarrow \frac{e^x}{e^x + 1} dx = \frac{y}{y + 1} dy \Rightarrow \ln(e^x + 1) = y - \ln|y + 1| + \ln c$$

$$\Rightarrow (e^x + 1) |y + 1| = ce^y$$

which can also be written as $(e^x + 1)(y + 1) = \pm ce^y$ 9. **A, B, D**

$$\vec{r}_1 \cdot \vec{r}_2 = |\vec{r}_1| \cdot |\vec{r}_2| \cos \theta \Rightarrow \cos \theta = 1. \text{ Hence, } \vec{r}_1 \text{ and } \vec{r}_2 \text{ are collinear vectors} \Rightarrow \frac{x}{a} = \frac{y}{b} = \frac{z}{c} = \frac{6}{5}$$

10. **A, C, D**

(A) Intersection point is mid-point of AB and CD

$$(B) 2\alpha + 6\beta + 15\gamma = 7$$

$$(2\hat{i} + 6\hat{j} + 3\hat{k}) \cdot (\alpha\hat{i} + \beta\hat{j} + 5\gamma\hat{k}) \leq 7 \times \sqrt{\alpha^2 + \beta^2 + 25\gamma^2}$$

$$(2\alpha + 6\beta + 15\gamma) \leq 7\sqrt{\alpha^2 + \beta^2 + 25\gamma^2}$$

$$\alpha^2 + \beta^2 + 25\gamma^2 \geq 1$$

(C) Triangle is equilateral

(D) Intersection points are (1, 2, 3) and (2, -3, 1)

11. **A, C, D**

$$\text{We have } P(A \cup B) \geq \max \{P(A), P(B)\} = \frac{4}{5}.$$

$$\text{Next } P(A \cap B) = P(A) + P(B) - P(A \cup B) \geq P(A) + P(B) - 1 = \frac{11}{20}.$$

$$\text{And } P(A \cap B) \leq \min \{P(A), P(B)\} = \frac{3}{4} \Rightarrow \frac{3}{4} \leq P(A \cap B) \leq \frac{11}{20}$$

$$P(A \cap B') = P(A) - P(A \cap B) \leq \frac{3}{4} - \frac{11}{20} = \frac{1}{5}$$

$$P(A' \cap B) = P(B) - P(A \cap B) \leq \frac{4}{5} - \frac{11}{20} = \frac{1}{4}$$

$$P(A \cup B) - \frac{1}{2} \leq P(A' \cap B) \leq P(A \cup B) - \frac{1}{6} \Rightarrow \frac{1}{6} \leq P(A \cap B') \leq \frac{1}{2}.$$

12. **B, D**

$$P(X_1) = \frac{1}{2}, P(X_2) = \frac{1}{4}, P(X_3) = \frac{1}{4}$$

$$P(X) = P(X_1 \cap X_2 \cap X_3^c) + P(X_1 \cap X_2^c \cap X_3) + P(X_1^c \cap X_2 \cap X_3) + P(X_1 \cap X_2 \cap X_3) = \frac{1}{4}$$

$$(A) P(X_1^c | X) = \frac{P(X \cap X_1^c)}{P(X)} = \frac{\frac{1}{32}}{\frac{1}{4}} = \frac{1}{8}$$

$$(B) P[\text{exactly two engines of the ship are functioning} | X] = \frac{\frac{7}{32}}{\frac{1}{4}} = \frac{7}{8}$$

$$(C) P\left(\frac{X}{X_2}\right) = \frac{\frac{5}{32}}{\frac{1}{4}} = \frac{5}{8}$$

$$(D) P\left(\frac{X}{X_1}\right) = \frac{\frac{7}{32}}{\frac{1}{2}} = \frac{7}{16}$$

Part - B

1. 11.00

$$P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$$

$$|\text{Adj } A| = |A|^2$$

$$|\text{Adj } A| = 16$$

$$1(12 - 12) - \alpha(4 - 6) + 3(4 - 6) = 16.$$

$$2\alpha - 6 = 16.$$

$$2\alpha = 22.$$

$$\alpha = 11.$$

2. 4.00

For least area $\lambda = 0$

$$\text{So; } A_{\min} = 2 \int_0^4 (12 - (x^2 - 4)) dx = \frac{256}{3}$$

3. 0.83

$$f'(x) = 2x + 1 \Rightarrow f(x) = x^2 + x + c$$

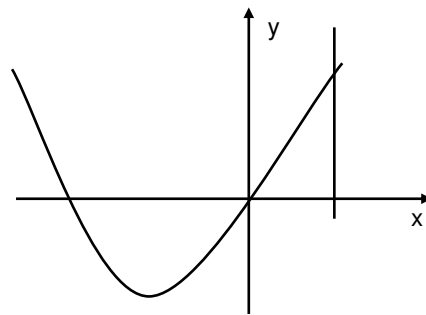
since curve passes through (1, 2), then

$$f(1) = 2 \Rightarrow c = 0$$

$$\text{hence } f(x) = x^2 + x$$

$$\text{required area (A)} = \int_0^1 (x^2 + x) dx$$

$$\Rightarrow A = \left(\frac{x^3}{3} + \frac{x^2}{2} \right)_0^1 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}.$$



4. 3.00

$$\text{Given that } \frac{dy}{dx} = 3x \Rightarrow y = \frac{3x^2}{2} + c$$

Since it is passing through $(-1, -3)$

$$\text{Then } c = -\frac{9}{2}$$

$$\therefore y = \frac{3x^2}{2} - \frac{9}{2} \Rightarrow 2y = 3(x^2 - 3) \therefore \lambda = 3.$$

5. 25.00

$$\text{Coefficient of } x^6 \text{ in } (x^5 + x^4 + \dots + 1)^3$$

$$\Rightarrow \text{Coefficient } x^6 \text{ in } (x^6 - 1)^3 (x - 1)^{-3}$$

$$\Rightarrow 28 - 3 = 25, R = 25$$

6. 5.00

$$\begin{vmatrix} a & 1 & 1 \\ 1 & a & 1 \\ 1 & 1 & a \end{vmatrix} = 0$$

$$a = 1, a = -2$$

for $a = 1$, it has infinite solutions so $a = -2$.